SKYFORCE

Drone Design & MATLAB Simulation
Consolidated Change Log

Purpose. This report consolidates the design decisions, simulation logic changes, sizing assumptions, outputs, and remaining validation limits discussed for the SKYFORCE interceptor. It is intended as a common reference for the team, guide, MATLAB work, and CAD coordination.

Snapshot date: 10 October 2026 Team: Team 08, Anti-Drone Interceptor, Senior Track

Current model type: 2D point-mass, illustrative scenario simulation. The outputs support geometric trajectory and state-logic discussion; they do not establish real flight approval, physical net capture, parachute deployment safety, or measured propulsion performance.

Executive summary

Topic	Current decision
Boundary / launch gate	Outer 2,000 m ring initiates track assessment; inner 1,000 m circle is the illustrative intercept-before-entry boundary. Launch recommendation depends on predicted track and available time, not just ring entry.
Vehicle baseline	18.00 kg nominal; 23.50 kg post-capture; U8 Lite KV100 / G28x9.2 CF 28-inch / Alpha 60A 12S.
Capture / release logic	5-15 m is an assumed geometric capture-candidate interval. Inside a 120 m safe-point radius: immediate-release state; outside: tow to nearest safe point; target mass >5 kg at candidate: parachute-request flag and no tow.
Constraints / energy	25 m/s speed cap; 4.5 m/s^2 model acceleration cap; tow-speed target 5 m/s and tow phase limited to 60 s. Sizing ceilings: 207 Wh no-tow and 253 Wh tow.
Outstanding proof	Run the latest MATLAB code and inspect the CSVs. Confirm every phase respects the acceleration cap; current complete-mission energy values are estimates unless continuously integrated.

How to use this document

Use the “current baseline” sections when editing MATLAB and drafting the design report. Historic values marked withdrawn must not be mixed with the revised configuration. Items marked assumed, provisional, or not modelled must retain those labels in reports and presentations.

1. Design baseline: what changed

The largest correction was replacing the first-pass propulsion and energy assumptions with a single revised baseline. The earlier sizing pass and the current guide decisions must not be blended.

Current authoritative baseline

Parameter	Current value	Status / note
Configuration	Hexarotor, electric; intercept-only mission model plus a separate towing branch	2D simulation is illustrative; not autopilot code.
Motor	T-Motor U8 Lite KV100; 238 g each incl. cable	Published mass figure used in the decision record; verify exact supplier part.
Propeller	T-Motor G28x9.2 CF, 28 inch	Matches the cited motor test configuration; actual propeller mass still needs confirmation.
ESC	T-Motor Alpha 60A, 12S	Exact ESC mass remains to be confirmed.
Max static thrust	7.3 kgf per motor; six motors	Published/static reference; not a forward-flight thrust curve.
Nominal mass / post-capture	18.00 kg / 23.50 kg	Use consistently in the simulation.
Arm / frame span	0.800 m centre-to-motor; 2.31 m frame span	Dimensions reported in the guide decision record; still coordinate with final prop clearance and CAD envelope.
Battery	12S, 8 Ah; 44.4 V nominal; estimated 1.97 kg	1.97 kg is estimated from generic energy density, not yet a selected product datasheet.

Mass ledger

Ledger item	Mass	Meaning
Listed component subtotal	**14.78 kg**	Subtotal reported in the revised resolution note.
Integration / growth allowance	**3.22 kg**	Named, visible allowance; not yet closed against real parts.
Nominal mass	**18.00 kg**	14.78 + 3.22 = 18.00 kg.
Post-capture mass	**23.50 kg**	Use after capture for the modelled tow branch; verify exact included loads.

Before literal CAD freeze: confirm the G28x9.2 propeller mass, Alpha 60A ESC mass, specific 12S 8 Ah battery product/mass, and net launcher + CO2 cartridge + net + parachute package envelope. The 3.22 kg allowance closes arithmetic provisionally; it is not a substitute for a final itemised bill of materials.

2. Retired assumptions and propulsion limits

Withdrawn first-pass values

Old value	Decision
T-Motor U8 II KV100 with 22-inch propeller	Withdrawn. Do not treat as the current configuration.
9.2 kgf max thrust per motor; ~41 A; ~1.965 kW per motor	Withdrawn. The guide resolution says these figures did not match a published datasheet for the claimed combination.
4.0 kW sprint power	Withdrawn. Current revised sprint-power assumption is 3.61 kW.
236 Wh intercept-only energy	Withdrawn from the revised baseline. Use 207 Wh no-tow and 253 Wh with tow as sizing ceilings.
18.06 kg mass with a 2.50 kg unexplained gap	Replaced by 18.00 kg nominal and 3.22 kg named allowance in a revised ledger (14.78 kg subtotal).

Thrust and model limits

Check	Nominal 18 kg	Post-capture 23.5 kg
Weight	176.6 N	230.5 N
Total static thrust (six x 7.3 kgf)	429.7 N	429.7 N
Static thrust-to-weight ratio	2.43	1.86
Static-available acceleration estimate	21.8 m/s^2	15.4 m/s^2
Static-based thrust margin at 25 m/s	2.27x	1.79x
Margin at 25 m/s after assumed 30% thrust derating	**1.59x**	**1.25x**

The 15%, 20%, and 30% forward-flight derating figures are sensitivity assumptions, not measured data. The 30% case is the conservative headline case approved for the hackathon-level report. There is no measured thrust-versus-airspeed curve for the selected propeller in the project record.

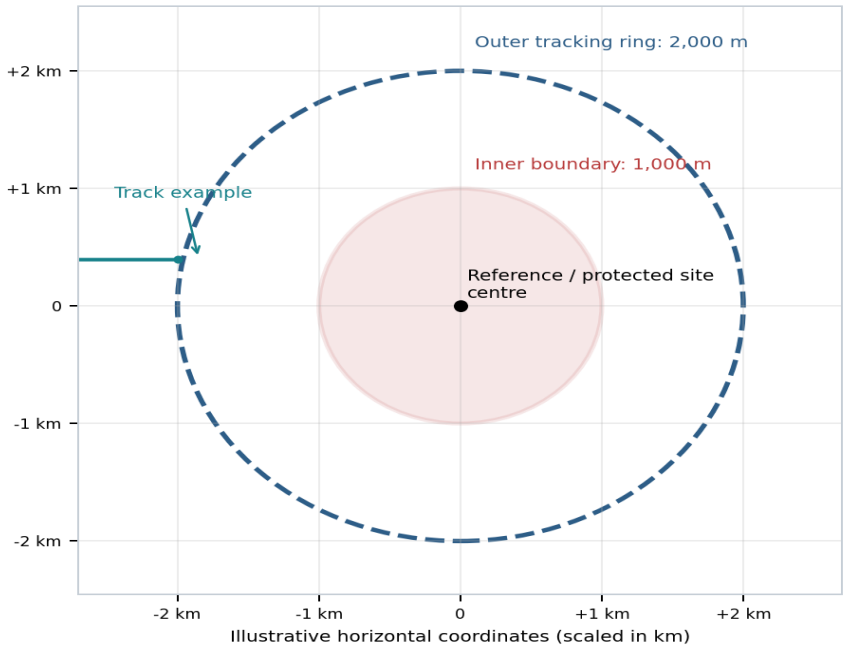
The 25 m/s speed cap and 4.5 m/s^2 acceleration cap

The 25 m/s value remains the simulation speed cap. The 4.5 m/s^2 cap is retained as a deliberately conservative control-smoothness / structural-load assumption. The static estimates show theoretical margin, but do not independently establish actual aircraft performance in forward flight. In the code, the acceleration limiter must be applied uniformly in every phase, including transitions.

> Interpretation: a plotted maximum speed of 25 m/s means the simulation kept the state below its configured cap. It is not, by itself, proof of measured flight capability. Likewise, 4.5 m/s^2 is a model cap whose applied values must be checked in the sample-by-sample CSV.

3. Two-boundary launch gate and launch location

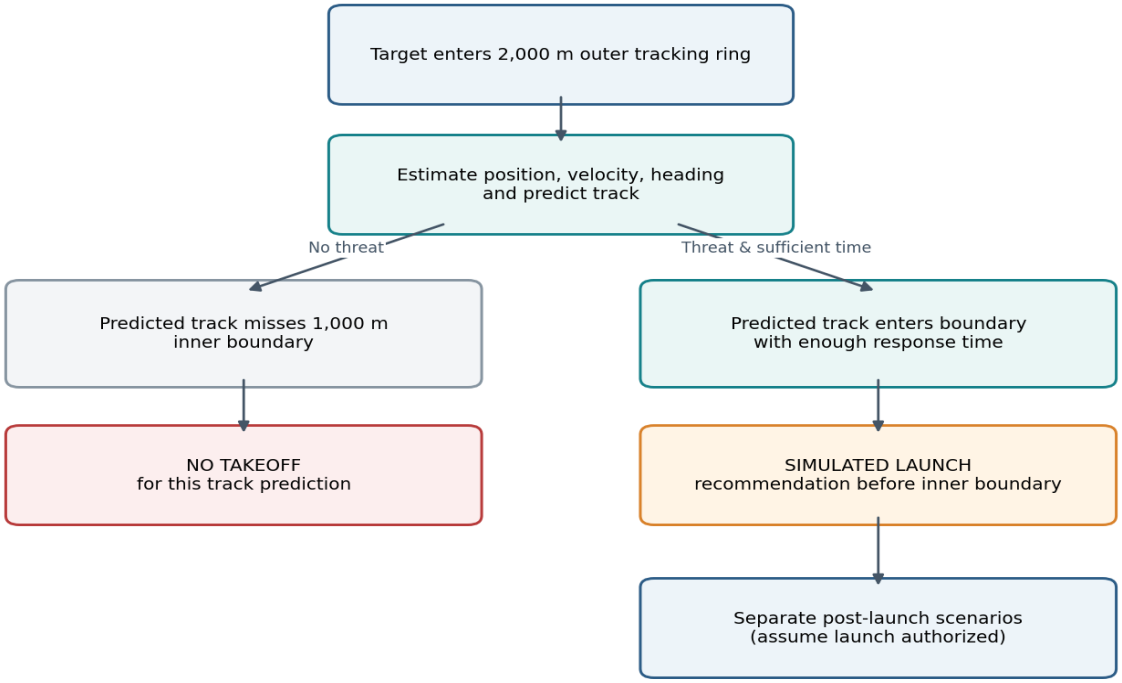
The most recent requirement uses two concentric radii centred on an illustrative protected-site reference point. These are scenario / decision boundaries, not the aircraft's physical dimensions and not approved operational limits.



Boundary definitions

Boundary / parameter	Current value	Decision meaning
Outer tracking ring	2,000 m radius	Target enters this ring; begin/record track assessment and estimate position, speed, heading and predicted path.
Inner protected boundary	1,000 m radius	The target should be intercepted before crossing this boundary, if a launch is authorised by the predicted path/time logic.
Designated safe-point radius	120 m around each safe-point centre	Used after a geometric capture candidate to decide immediate-release state versus tow-to-safe-point state. Separate concept from the 1,000 m inner boundary.
Earlier 300 m circle	Earlier placeholder	Superseded in the latest mission plots/config by the requested 1,000 m inner boundary. Do not mix labels from old figures with the current setting.

Launch gate flow



Offline scenario logic only - no real radar, autopilot, launcher or parachute commands

What “fixed at one point” means here: the interceptor has a fixed initial pad/base coordinate while it is waiting for the launch decision. The separate gate demonstration leaves it stationary and returns a simulated decision. The four post-launch mission trajectories are separate cases that assume launch was already authorised; the present code is not yet one continuous gate-to-landing simulation.

Gate timing rule and illustrative cases

Case	Predicted result
Track enters outer ring, predicted path intersects inner circle, and time available exceeds 33.8 s	Simulated launch recommendation before inner-boundary entry.
Track enters outer ring but closest approach remains outside 1,000 m	NO TAKEOFF for this predicted track.
Track is detected but time to inner entry is <= 33.8 s	NO TAKEOFF: too late under current illustrative response-time assumption.

The 33.8 s test is the current illustrative sum of 1.8 s reaction delay + 30 s estimated interception time + 2 s safety margin. A real integrated model should calculate time-to-intercept per target state; this fixed comparison is a hackathon demonstration, not a certified launch rule.

4. Launch timing, guidance, and mission cases

Current launch-gate timing assumptions

Input	Current assumption	Purpose
Reaction delay	1.8 s	Stacked estimate for track update, ground processing / planning, radio link and vehicle spin-up; not measured.
Estimated interception time for gate example	30 s	Illustrative gate calculation input. Actual case-specific intercept time should come from a flight model.
Additional time margin	2 s	Additional illustrative safety margin.
Total response estimate used by gate	33.8 s	1.8 + 30 + 2 seconds. If time from outer-ring track acquisition to inner-boundary entry is not greater than this, gate reports no takeoff.

The launch-gate demo includes three synthetic examples: (1) predicted inner-boundary entry with enough time, which creates a simulated launch recommendation; (2) target stays outside / misses the 1,000 m circle, so no takeoff; and (3) target is detected too late, so no takeoff is recommended. The logic uses constant-velocity track prediction and is not a radar simulation.

Post-launch scenarios in the MATLAB file

Scenario	Inputs / logic	Expected state outcome
1. Attack / slower target	Target vector [14, 5] m/s; speed about 14.87 m/s; attack/tail-chase geometry.	Geometric candidate in 5-15 m window; if inside safe area, immediate-release state.
2. Defense / faster target	Target vector [27, 0] m/s; target faster than 25 m/s cap; move to lead point and wait for target.	Geometric candidate; demonstrates lead-point idea, not physical capture.
3. Tow required	Target vector [12, 0] m/s; candidate outside all 120 m safe-point radii.	Tow to nearest safe point; tow duration target <= 60 s, speed setpoint 5 m/s.
4. Abort	Target vector [-20, 0] m/s; abort if hard boundary crossed or time-to-boundary <= estimated intercept time + 2 s margin.	Pre-launch or in-simulation abort; energy compliance is not scored as a mission pass if no launch occurred.
5. PN diagnostic	Initial interceptor speed 20 m/s, target vector [0,12] m/s; deliberate heading/LOS offset; PN gain N=3, blend range 600 m.	Additional guidance diagnostic only; not a full mission scenario.

Important geometry note: launch gate uses a 2,000 m outer radius and 1,000 m inner radius about a site centre. Capture-release logic uses a separate 120 m radius around each safe-point centre. These radii have different functions.

5. Capture-candidate, release/tow, and overweight policy

The 5-15 m interval is the assumed geometric engagement window. It is a separation interval between interceptor and target, not a time window and not the 1,000 m boundary.

Required report wording: use “geometric capture candidate” for simulation-derived events. Do not write “successful capture” or “capture achieved” based solely on this point-mass model.

Decision after a geometric candidate

Condition at candidate	Modelled decision	What it does not prove
Target mass <= 5 kg and candidate within any safe-point radius of 120 m	Immediate-release state at candidate / inside designated safe area.	Does not model release mechanics or landing dynamics.
Target mass <= 5 kg and candidate outside all safe-point radii	Tow to nearest safe-point centre using a low-speed waypoint controller; max tow phase 60 s.	Does not validate loaded flight, physical attachment, or actual flight controllability.
Target mass > 5 kg at candidate	Set simulation-only parachute-request flag; disable towing. This overweight policy overrides safe-point tow choice.	Not a real deployment command and not proof the existing parachute can safely recover the load.

Overweight target rule

The team requested that a target exceeding 5 kg should not lead to towing. The simulation therefore does not reject launch solely because an overweight mass estimate is present; it follows the trajectory/time launch gate. If an overweight target reaches a geometric candidate event, the simulation requests parachute recovery and disables towing.

Critical physical-design caveat: the existing 2.5 m parachute sizing calculation assumes a 5.5 kg suspended load (5 kg target plus an assumed 0.5 kg net/tether). A 6 kg target would exceed that load case. The MATLAB flag only represents the intended state logic; a suitably rated parachute, load path, opening-shock and landing check would be needed before claiming safe recovery.

Tow transition correction

The previously observed towing acceleration spike (about 222.89 m/s^2) was caused by an abrupt speed change at the capture-to-tow mode transition. The documented correction is to synchronise the mass update at capture, start the tow velocity target from the actual capture velocity, ramp toward a 5 m/s tow setpoint, and apply the same 4.5 m/s^2 acceleration limiter in every mode.

Hand check for a conservative 25 to 5 m/s transition: dV = 20 m/s; at 4.5 m/s^2, ramp time = 20/4.5 = 4.44 s. At constant deceleration, average speed is 15 m/s and transition distance is about 66.7 m. This hand calculation shows the intended ramp logic; it does not replace checking the logged acceleration in a rerun.

6. Power, battery, and mission-energy accounting

Current sizing baseline

Quantity	Value	Interpretation
Estimated hover power	1.56 kW	Revised propulsion baseline; estimated from the selected configuration.
Estimated sprint power at 25 m/s	3.61 kW	Replaces withdrawn 4.0 kW first-pass figure.
No-tow sizing ceiling	207 Wh	120 s sprint + 120 s hover-equivalent profile, plus 20% demand-side margin.
Tow sizing ceiling	253 Wh	207 Wh base plus a separately margined 60 s loaded tow addition, per the selected convention.
Battery nominal energy	355.2 Wh	44.4 V x 8 Ah.
Energy counted usable	284.16 Wh	80% of nominal pack energy counted as usable; a conservative state-of-charge floor.
Additional phase buffer	9.7 Wh no-tow; 18.8 Wh tow	Estimated allowance for launch spin-up, post-release/landing descent, and avionics/radio. Not a measured draw.

Energy estimates and version history

Figure	Status
236 Wh no-tow	Withdrawn first-pass value. Do not use as current design ceiling.
207 Wh no-tow / 253 Wh with tow	Current authoritative sizing ceilings.
39.55 Wh attack / 41.44 Wh defense / 103.53 Wh tow	Guide Decision Record estimates using its then-current CSV phase-energy values plus the agreed buffers.
105.11 Wh tow	Later provisional estimate in the towing-transition correction note: prior 84.73 Wh phase value + 1.58 Wh transition correction + 18.8 Wh buffer. Explicitly not a corrected MATLAB run.
Current rerun value	Must come from the latest corrected MATLAB CSV. Do not mix phase-energy values from different runs.

Energy-reporting rule: display (1) energy used in simulated phases, (2) the added-phase buffer, (3) the resulting complete-mission estimate, and (4) the 207/253 Wh ceiling. Until one continuous launch-to-landing integration is implemented, label complete-mission values as estimates. The 20% demand margin and 80% usable-capacity convention are two distinct, intentionally stacked margins.

7. MATLAB outputs, CSV fields, and verification checks

The latest MATLAB file is designed to generate four main post-launch cases, a terminal-approach summary, a PN guidance diagnostic, an offline boundary-gate demonstration, and an overweight-target decision demonstration.

Expected output files

Output	What it is for
skyforce_simulation_summary.csv	One row per scenario: mode, launch-gate status, target mass, capture-candidate state, safe-area decision, tow status, speed / acceleration checks, energy estimates and limits.
skyforce_trajectory_export.csv	Sample-by-sample state log: time, phase, interceptor and target position/velocity, line of sight, range rate, closing speed, requested/applied acceleration, active limiter, power and cumulative simulated energy.
skyforce_propulsion_sensitivity.csv	Available thrust and thrust-margin estimates for assumed 15%, 20% and 30% forward-flight derating cases.
skyforce_boundary_gate_summary.csv	Synthetic two-boundary scenarios: track entry, closest approach, time to inner-boundary entry, response-time comparison and simulated gate decision.
skyforce_overweight_policy_summary.csv	Mass limit, parachute-request flag, towing allowed/not allowed, deployment modelled and recovery validated fields.

Minimum verification checklist after running MATLAB

- 1. Boundary gate: confirm 2,000 m and 1,000 m are the radii actually reported by the generated gate CSV and plot. Check the three example decisions: enough time / inner boundary missed / too late.
- 2. Capture candidate: all candidate ranges should lie in 5-15 m; post-capture joined positions must not overwrite the pre-capture closest-range metric.
- 3. Safe area: immediate release only when candidate is within a 120 m safe-point radius; otherwise tow, except overweight targets use the parachute-request/no-tow state.
- 4. Acceleration: inspect the maximum applied acceleration for every phase, especially the exact sample when mass and mode change. Require <= 4.5 m/s^2 for simulated applied acceleration; no direct velocity clipping.
- 5. Closing speed: independently recompute from exported relative velocity and line-of-sight unit vector using Vclose = -range rate.
- 6. Energy: ensure the CSV run values, buffers, complete-mission estimates, and sizing ceiling all come from one run / one explicitly stated set of assumptions.
- 7. Overweight case: verify mass >5 kg sets parachute-request flag and towing-disabled flag after a geometric candidate; do not label physical deployment successful.
- 8. File saving: confirm MATLAB printed the actual paths where all CSV files were saved; filenames may be timestamped if a file was locked.

Test status: the code has evolved through several edits. A PDF specification or hand calculation does not prove that a particular MATLAB run passed. Use the latest local run and its exported CSV as the evidence, and keep old CSVs/screenshots clearly versioned.

8. What is demonstrated, what remains open, and CAD implications

Scope and confidence

Item	Current status	Next confirmation / evidence
Launch gate / 2 radii	Illustrative constant-velocity track/time logic; no hardware command.	Verify model outputs; actual site limits remain configurable assumptions.
Attack / defense geometry	Shows whether trajectories reach the assumed 5-15 m interval.	Closing-speed check from exported vectors; PN diagnostic should show meaningful correction.
Tow transition	Velocity-ramp fix is designed into the latest code; prior output showed a 222.89 m/s^2 spike.	Rerun; confirm every applied acceleration sample <=4.5 m/s^2.
Full mission energy	Phase-energy plus buffer estimate; not continuous integration.	Use current CSV consistently; continuous launch-to-landing integration recommended if time permits.
Forward-flight propulsion	15-30% derating is an assumed sensitivity range; 30% case is headline.	Measured thrust-vs-airspeed curve would fully close the technical uncertainty.
Mass ledger / CAD	18.0 kg mass includes a provisional 3.22 kg allowance.	Confirm actual propeller, ESC, battery masses and payload envelope before literal final freeze.
Net / parachute	Capture and recovery are not modelled; >5 kg path only sets a simulation flag.	Confirm physical net canister / cartridge / canopy mass and package envelope; load rating and deployment dynamics unvalidated.
Site boundaries	2,000 m, 1,000 m and 120 m circles are illustrative scenario values.	Actual site-specific boundaries not needed for the demo but must be approved before any real operation.

Would the logic apply to fixed-wing or VTOL?

Boundary / track-prediction logic: conceptually transferable. Estimating target position, velocity and heading, predicting whether a track enters a protected region, and checking available response time are not limited to multirotors.

Current interceptor dynamics: not directly transferable as-is. The current 2D point-mass controller assumes multirotor-like speed control, acceleration limiting, the ability to slow to a low-speed towing target, and waypoint/hold behaviour.

Fixed-wing aircraft: would need a model for minimum safe airspeed/stall margin, lift and drag, turn radius/bank limits, climb performance and the fact that it cannot hover or stop at a lead point.

Hybrid VTOL: would need separate vertical-hover, transition, and wing-borne cruise dynamics, with mode-specific power and control limits.

CAD-freeze decisions: arm geometry/span can proceed provisionally from the revised 28-inch propeller baseline, but confirm actual propeller and ESC masses, the selected battery product and mass, and the net/CO2/parachute package envelope. The site radii and PN tuning are software / scenario settings, not airframe geometry. Physical capture and recovery performance are intentionally out of scope for this preliminary deliverable.

Source record for this consolidation

This change log consolidates the user-provided project materials and decisions: SKYFORCE - Interceptor Sizing Calculations; SKYFORCE - MATLAB Intercept Simulation: What It Should Show; Resolution of Simulation Issues; Final Resolution of Remaining Simulation Issues; Guide Decision Record; Towing Transition Correction and Revised Energy Estimate; and the latest MATLAB source file skyforce_intercept_simulation_updated.m. If a new run changes scenario energy or acceleration results, the current CSV export should supersede historical screenshot values, with the change explained.